Trig Practice Test 3

Trig equations · Bearings · Law of Sines · Law of Cosines — 20 questions · Answer key on pages 9–10

Name _____ Date _____ Score _____ / 20

Part A — Higher-Degree Trig Equations

Solve on [0, 2π). Let u = sin x or u = cos x, factor, and solve each factor.

1. $2\sin^2 x - 3\sin x + 1 = 0$

$\frac{3 \pm \sqrt{9 - 4(2)(1)}}{4}$

$\frac{3 \pm \sqrt{9 - 8}}{4}$

$\frac{3 \pm \sqrt{1}}{4}$

$\sin x = \frac{3+1}{4}$

$\sin x = 1$

$x = \frac{\pi}{2}$

$\sin x = \frac{3-1}{4}$

$\sin x = \frac{2}{4}$

$\sin x = \frac{1}{2}$

$x = \frac{\pi}{6}, \frac{5\pi}{6}$

2. $2\cos^2 x + \cos x - 1 = 0$

$\frac{-1 \pm \sqrt{1 - 4(2)(-1)}}{4}$

$\frac{-1 \pm \sqrt{1 + 8}}{4}$

$\frac{-1 \pm \sqrt{9}}{4}$

$\frac{-1 \pm 3}{4}$

$\cos x = \frac{-1+3}{4}$

$\cos x = \frac{2}{4}$

$\cos x = \frac{1}{2}$

$x = \frac{\pi}{3}, \frac{5\pi}{3}$

$\cos x = \frac{-1-3}{4}$

$\cos x = \frac{-4}{4}$

$\cos x = -1$

$x = \pi$

3. $2\cos^2 x + 5\cos x + 3 = 0$

$$\frac{-5 \pm \sqrt{25 - 4(2)(3)}}{4}$$

$$\frac{-5 \pm \sqrt{1}}{4}$$

$$\frac{-5 \pm 1}{4}$$

$\cos x = \frac{-5+1}{4}$

$\cos x = \frac{-4}{4}$

$\cos x = -1$

$x = \pi$

$\cos x = \frac{-5-1}{4}$

$\cos x = \frac{-6}{4}$

$\cos x = -\frac{3}{2}$ X

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4. Write the **general solution** (all real x):
$2\sin^2 x - \sin x - 1 = 0$

$$\frac{1 \pm \sqrt{1 - 4(2)(-1)}}{4}$$

$$\frac{1 \pm \sqrt{9}}{4}$$

$$\frac{1 \pm 3}{4}$$

$\sin x = \frac{1+3}{4}$

$\sin x = \frac{4}{4}$

$\sin x = 1$

$x = \frac{\pi}{2}$

$\sin x = \frac{1-3}{4}$

$\sin x = \frac{-2}{4}$

$\sin x = -\frac{1}{2}$

$x = \frac{7\pi}{6}, \frac{11\pi}{6}$

$x = \frac{\pi}{2} + 2n\pi$ & $\frac{7\pi}{6} + 2n\pi$ & $\frac{11\pi}{6} + 2n\pi$

Part B — Use Identities to Solve

Solve on [0, 2π). sin 2x = 2 sin x cos x. Move everything to one side and factor.

5. sin 2x = sin x

$2\sin x \cdot \cos x = \sin x$

$2\sin x \cdot \cos x - \sin x = 0$

$\sin x(2\cos x - 1)$

$\sin x = 0$

$x = 0, 2\pi$

$2\cos x - 1 = 0$

$2\cos x = 1$

$\cos x = \frac{1}{2}$

$x = \frac{\pi}{3}, \frac{5\pi}{3}$

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6. cos x = −sin x

$-\frac{\sin x}{\cos x} = 1$

$-\tan x = 1$

$\tan x = -1$

$x = \frac{3\pi}{4}, \frac{7\pi}{4}$

7. $\sin 2x + \sqrt{3}\cos x = 0$

$2\sin x \cdot \cos x + \sqrt{3}\cos x = 0$

$\cos(2\sin x + \sqrt{3}) = 0$

$\cos x = 0$

$x = \frac{\pi}{2}, \frac{3\pi}{2}$

$2\sin x + \sqrt{3} = 0$

$2\sin x = -\sqrt{3}$

$\sin x = -\frac{\sqrt{3}}{2}$

$x = \frac{4\pi}{3}, \frac{5\pi}{3}$

8. $\sqrt{3}\sin x = \cos x$

$\sqrt{3}\left(\frac{\sin x}{\cos x}\right) = 1$

$\sqrt{3}\tan x = 1$

$\tan x = \frac{1}{\sqrt{3}}$

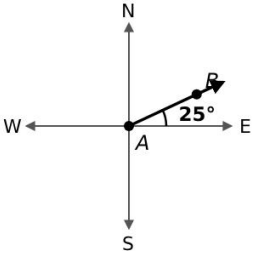
$\tan x = \frac{\sqrt{3}}{3}$

$x = \frac{\pi}{6}, \frac{7\pi}{6}$

Part C — Bearings

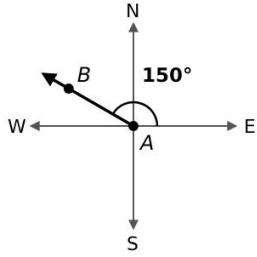
Find the bearing from A to B: N or S, then the acute angle from the north–south line, then E or W.

9. Bearing from A to B?



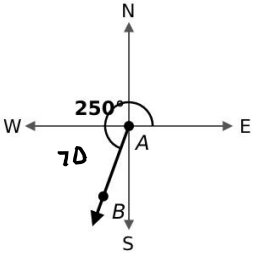
N 65 E

10. Bearing from A to B?



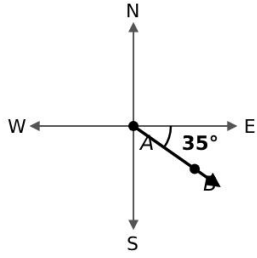
N 60 W

11. Bearing from A to B?



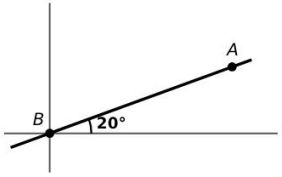
S 20 W

12. Bearing from A to B?



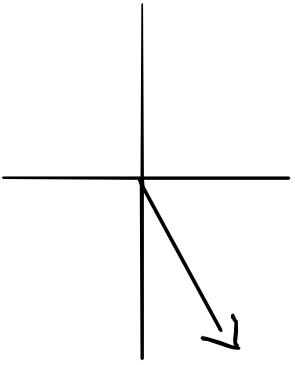
S 55 E

13. Find the bearing from A to B. (Hint: put a compass at A.)



S 70 W

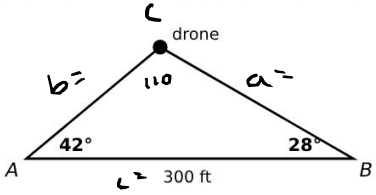
14. A ship leaves port heading **S35°E**. Describe in words which way it is traveling, and sketch it on a compass.



Part D — Law of Sines

a / sin A = b / sin B = c / sin C

15. A drone hovers directly above line AB and is observed at the same moment from A and B, which are 300 ft apart. The angle of elevation from A is 42° and from B is 28°. Find the distance from each location to the drone. Round to the nearest foot.



$\frac{300}{\sin 110} = \frac{a}{\sin 42}$ $\quad a = \frac{300 \sin 42}{\sin 110}$

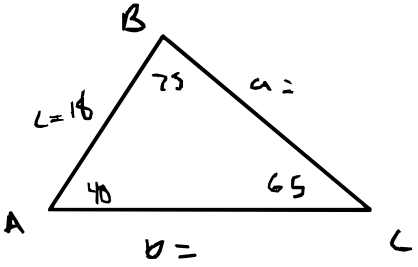
$a = 214$

$\frac{b}{\sin 28} = \frac{300}{\sin 110}$

$b = \frac{300 \sin 28}{\sin 110}$

$b = 150$

16. In △ABC, A = 40°, B = 75°, and c = 18. Solve the triangle. Round to 1 decimal place.



$\frac{18}{\sin 65} = \frac{a}{\sin 40}$ $\quad \frac{18}{\sin 65} = \frac{b}{\sin 75}$

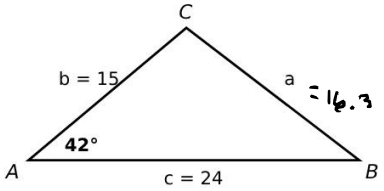
$a = \frac{18 \sin 40}{\sin 65}$ $\quad b = \frac{18 \sin 75}{\sin 65}$

$a = 12.7$ $\quad b = 19.2$

Part E — Law of Cosines

$a^2 = b^2 + c^2 - 2bc\cos A$ (gets sides) · $\cos A = (b^2 + c^2 - a^2) / 2bc$ (gets angles)

17. Solve the triangle: b = 15, c = 24, A = 42°. Round to 1 decimal place.



$a^2 = 24^2 + 15^2 - 2 \cdot 15 \cdot 24 \cdot \cos 42$

$a^2 = 265.9$ $a = 16.3$

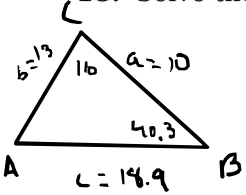
$\cos B = \dfrac{16.3^2 + 24^2 - 15^2}{2 \cdot 16.3 \cdot 24}$

$C = 180 - 38 - 42$

$C = 100$

$\cos B = \dfrac{616.7}{782.4}$ $B = \cos^{-1}\left(\dfrac{616.7}{782.4}\right)$ $B = 38°$

18. Solve the triangle: a = 10, b = 13, C = 110°. Round to 1 decimal place.



$c^2 = 10^2 + 13^2 - 2 \cdot 13 \cdot 10 \cdot \cos 110$

$c^2 = 357.9$ $c = 18.9$

$\cos B = \dfrac{10^2 + 18.9^2 - 13^2}{2 \cdot 10 \cdot 18.9}$

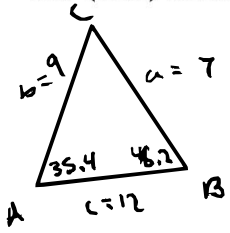
$\cos B = \dfrac{288.2}{378}$

$B = \cos^{-1}\left(\dfrac{288.2}{378}\right)$ $B = 40.3°$

$A = 180 - 40.3 - 110$

$A = 29.7°$

19. (SSS) Solve the triangle: a = 7, b = 9, c = 12. Round to 1 decimal place.



$\cos A = \dfrac{9^2 + 12^2 - 7^2}{2 \cdot 9 \cdot 12}$ $\cos B = \dfrac{7^2 + 12^2 - 9^2}{2 \cdot 7 \cdot 12}$

$A = \cos^{-1}\left(\dfrac{176}{216}\right)$ $B = \cos^{-1}\left(\dfrac{112}{168}\right)$

$C = 180 - 35.4 - 48.2$

$C = 96.4$

$A = 35.4°$ $B = 48.2°$

20. Law of Sines or Law of Cosines first?

(a) two sides + included angle (b) two angles + a side (c) three sides

Answer Key

Worked solutions

PART A

1. $2u^2 - 3u + 1 = 0 \rightarrow (2u - 1)(u - 1) = 0$
$\sin x = ½ \rightarrow \pi/6, 5\pi/6$ $\sin x = 1 \rightarrow \pi/2$
x = π/6, π/2, 5π/6

2. $2u^2 + u - 1 = 0 \rightarrow (2u - 1)(u + 1) = 0$
$\cos x = ½ \rightarrow \pi/3, 5\pi/3$ $\cos x = -1 \rightarrow \pi$
x = π/3, π, 5π/3

3. $2u^2 + 5u + 3 = 0 \rightarrow (2u + 3)(u + 1) = 0$
$\cos x = -3/2 \rightarrow$ no solution (cos must be between −1 and 1)
$\cos x = -1 \rightarrow \pi$
x = π

4. $(2u + 1)(u - 1) = 0 \rightarrow \sin x = -½$ or $\sin x = 1$
$\sin x = -½ \rightarrow 7\pi/6, 11\pi/6$ $\sin x = 1 \rightarrow \pi/2$; add $2n\pi$ to each
x = π/2 + 2nπ, 7π/6 + 2nπ, 11π/6 + 2nπ

PART B

5. $2 \sin x \cos x - \sin x = 0 \rightarrow \sin x(2\cos x - 1) = 0$
$\sin x = 0 \rightarrow 0, \pi$ $\cos x = ½ \rightarrow \pi/3, 5\pi/3$
Don't divide both sides by sin x, or you lose 0 and π.
x = 0, π/3, π, 5π/3

6. Divide by $\cos x \rightarrow \tan x = -1$ ($\cos x = 0$ doesn't work, since $\sin x$ would be ±1)
tan is negative in QII and QIV, reference angle π/4
x = 3π/4, 7π/4

7. $2 \sin x \cos x + \sqrt{3} \cos x = 0 \rightarrow \cos x(2\sin x + \sqrt{3}) = 0$
$\cos x = 0 \rightarrow \pi/2, 3\pi/2$ $\sin x = -\sqrt{3}/2 \rightarrow 4\pi/3, 5\pi/3$
x = π/2, 4π/3, 3π/2, 5π/3

8. Divide by $\cos x \rightarrow \tan x = 1/\sqrt{3} = \sqrt{3}/3$
Reference angle π/6; tan is positive in QI and QIII
x = π/6, 7π/6

PART C

9. Ray is 25° above east → 90 − 25 = 65° from north, toward east
N65°E

10. 150° from east is 60° past north, toward west
N60°W

11. 250° from east is 70° past west → 90 − 70 = 20° from south, toward west
S20°W

12. 35° below east → 90 − 35 = 55° from south, toward east
S55°E

13. From A, B is 20° below due west → 90 − 20 = 70° from south, toward west
S70°W

14. Face due south, then turn 35° toward east
Southeast: 35° east of due south

PART D

15. Angle at drone: 180 − 42 − 28 = 110°
From A (side b, opposite B): b = 300 sin 28° / sin 110° ≈ 149.9
From B (side a, opposite A): a = 300 sin 42° / sin 110° ≈ 213.6
Watch labels: side a is the distance from B.
≈ 150 ft from A, ≈ 214 ft from B

16. C = 180 − 40 − 75 = 65°
a = 18 sin 40° / sin 65° ≈ 12.8 b = 18 sin 75° / sin 65° ≈ 19.2
C = 65°, a ≈ 12.8, b ≈ 19.2

PART E

17. $a^2 = 15^2 + 24^2 - 2(15)(24)\cos 42° \approx 801 - 535.1 = 265.9 \rightarrow a \approx 16.3$
Angle opposite shorter given side (15): $\cos B = (16.3^2 + 24^2 - 15^2)/(2 \cdot 16.3 \cdot 24) \approx 0.788 \rightarrow B \approx 38.0°$
C = 180 − 42 − 38.0 = 100.0° (obtuse, which is why you find the small angle first)
a ≈ 16.3, B ≈ 38.0°, C ≈ 100.0°

18. $c^2 = 100 + 169 - 260 \cos 110° = 269 + 88.9 = 357.9 \rightarrow c \approx 18.9$ (cos 110° is negative)
$\cos A = (13^2 + 18.9^2 - 10^2)/(2 \cdot 13 \cdot 18.9) \approx 0.868 \rightarrow A \approx 29.8°$
B = 180 − 110 − 29.8 = 40.2°
c ≈ 18.9, A ≈ 29.8°, B ≈ 40.2°

19. Largest angle first: $\cos C = (49 + 81 - 144)/(2 \cdot 7 \cdot 9) = -14/126 \rightarrow C \approx 96.4°$
$\cos A = (81 + 144 - 49)/(2 \cdot 9 \cdot 12) = 176/216 \rightarrow A \approx 35.4°$
B = 180 − 96.4 − 35.4 = 48.2°
A ≈ 35.4°, B ≈ 48.2°, C ≈ 96.4°

20. Law of Sines needs a known angle with its opposite side.
(a) Cosines (b) Sines (c) Cosines